Evaluation and comparison of local large models in news summarization

Hrvoje Vukić and Mirna Knez

Abstract

This paper presents an empirical analysis and evaluation of three locally deployed Large Language Models: Llama 3.2, Gemma 3.4B, and Qwen 3.4B. The models were tasked with automatically generating summaries of news articles from the news_summary dataset. Evaluation was conducted through a multidisciplinary approach encompassing computational efficiency in terms of execution time and peak memory consumption, traditional n-gram overlap metrics such as ROUGE, BLEU, and METEOR, as well as advanced semantic similarity measured via BERTScore and cosine similarity. Additionally, linguistic and structural parameters including lexical diversity, readability indices, keyword coverage, redundancy, and novelty levels were analyzed. The results demonstrate clear trade-offs between processing speed, linguistic creativity, and semantic precision across the evaluated models.

1.Introduction

Automatic text summarization is one of the foundational and most challenging tasks in the field of Natural Language Processing. With the emergence of modern opensource large language models, local execution on standard hardware has become feasible without the need to transmit data to external servers, thereby ensuring high data privacy and complete cost control. The objective of this study is to investigate how three popular local models of varying architectures perform when tasked with summarizing news articles to a strictly defined length. The focus of this work is not solely directed at execution speed, but rather at a deep investigation of the generated summaries' quality using a diverse suite of contemporary NLP evaluation metrics that span syntactic, semantic, and stylistic aspects of text.

2. Methodology and dataset

The experiment was conducted on a modified subset of the well-known news_summary dataset. For each news article, a corresponding human-written reference summary or headline was provided as the ground truth. Model inference was performed locally using the Ollama library, applying an identical prompt configuration with strict instructions to generate a summary consisting of exactly three sentences. The models under evaluation included Llama 3.2, Gemma 3 (4B), and Qwen 3 (4B). During the generation phase, computational performance metrics such as execution time and peak memory allocation were monitored, after which the generated texts were subjected to comprehensive statistical and linguistic evaluation utilizing custom Python scripts and Jupyter notebooks.

3.Results and conclusion

Based on the empirical data gathered, the mean values of all observed metrics were calculated for each individual model. In terms of computational efficiency, Llama 3.2 emerged as the fastest model by a substantial margin, achieving an average summary generation time of 5.683 seconds. Gemma 3 exhibited a noticeably slower execution time of 14.204 seconds, while Qwen 3 required 42.139 seconds per summary, rendering it less suitable for interactive or real-time applications. Python-level memory consumption, tracked via the tracemalloc library, remained consistent across all models at approximately 0.07 MB, which is a direct consequence of the core model inference running inside the background Ollama service while Python handles only the API requests and responses. When examining traditional text-matching metrics, Llama 3.2 achieved a definitive victory, securing the highest scores across ROUGE-1 (0.3476), ROUGE-2 (0.1781), ROUGE-L (0.2237), BLEU (0.0332), and METEOR (0.1585). This strong alignment suggests that Llama 3.2 excels at reproducing text structures that closely match human references, an observation further supported by its superior keyword coverage of 0.2478 and the highest recorded BERTScore of 0.8781. Despite Llama's dominance in n-gram overlap, advanced semantic and linguistic analysis reveals a different set of trends. The Qwen 3.4B model achieved the top performance in the semantic similarity category with an average score of 0.8052 and also demonstrated the highest lexical diversity at 0.8523, indicating a more sophisticated and less repetitive vocabulary profile. Concurrently, Gemma 3 stood out remarkably in terms of text novelty, achieving the highest index of 0.5338 alongside the lowest redundancy score of 0.1607. These metrics reveal that Gemma 3 successfully paraphrases source text using its own vocabulary rather than extracting whole sentences verbatim from the source article, thereby avoiding structural repetition. In terms of structural readability, Llama 3.2 led with a readability index of 33.0092, suggesting clearer sentence structures, though accompanied by a slightly elevated redundancy rate of 0.1948.

4.CONCLUSION

This research highlights a distinct trade-off when selecting a local large language model for automated text summarization tasks. If execution speed and strict adherence to a human reference baseline are the primary constraints, Llama 3.2 represents the optimal choice, delivering high structural keyword alignment and excellent text readability with minimal computational overhead. Conversely, if the application demands creative expression, greater linguistic novelty, and a deeper grasp of semantic context, Qwen 3 and Gemma 3 emerge as stronger alternatives. Qwen 3 excels in semantic consistency and vocabulary breadth, whereas Gemma 3 effectively minimizes textual redundancy and creates highly original, paraphrased abstracts, albeit at a higher computational time expense.